

Normalised Audio (16kHz PCM)																																																																																																			
Held in Volatile Memory Only																																																																																																			
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PASS ONE: Local In-Browser ASR Engine																																																																																																			
(WebGPU with Wasm Fallback)																																																																																																			
• Word-level timestamps (start / end)																																																																																																			
• Per-word confidence scores (0.0..1.0)																																																																																																			
• Channel/Acoustic Speaker Attribution																																																																																																			
• Low-confidence token escalation (<0.70)																																																																																																			
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Pass One Internal Transcript																																																																																																			
• NEVER leaves the adviser workstation																																																																																																			
• NEVER displayed as working case note																																																																																																			
• Lives solely in volatile RAM																																																																																																			
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PASS TWO: Acoustic Redaction Pipeline																																																																																																			
(Locates PII & zeros audio intervals)																																																																																																			

The Pass One transcript is not the working case note transcript. Whisper small/base models executed in-browser can exhibit minor acoustic errors, hallucinations, or phonetic variations. Its sole function is to provide word-level alignments and candidate token boundaries to feed the local PII extraction and acoustic silence-redaction engine (Phase 8). The adviser is explicitly presented with clear disclaimers preventing mistaking Pass One text for the final case note.

Pass One speech recognition executes 100% locally within a sandboxed Web Worker. All network primitives (fetch, XMLHttpRequest, WebSocket, EventSource) are stripped from the worker global scope. No segment, timestamp, or audio slice is transmitted outside the browser memory during Pass One.

WebAssembly (Wasm) Backend: Universal fallback for older hardware or virtualized enterprise environments without GPU compute support. Runs multithreaded SIMD operations. Realtime factor is $\sim 0.80\text{--}1.20\times$ realtime.

Consultation audio buffers, intermediate spectrograms, tokens, and transcripts are strictly classified as session data and are never cached or persisted.

2.3 Pre-Session Hardware Benchmarking & Adviser Warnings

Prior to commencing an advice interview, the application assesses the host device's GPU capabilities via `LocalAsrEngine.assessHardwareCapabilities()`. If WebGPU is unavailable, the adviser is alerted via the `<HardwareBenchmarkBanner />` so expectations on post-interview processing times are established before the consultation starts.

3. Speaker Attribution & Channel Disambiguation

Knowing whether a phrase originated from the client or the adviser significantly improves entity classification (e.g. distinguishing a client giving their National Insurance number from an adviser reciting an office reference) and downstream case note structure.

Webex Telephony (Exact Split):

When audio originates from the Webex stereo stream capture (Route = 'webex_telephony'), Channel 0 represents the adviser's local microphone and Channel 1 represents the client's telephone line. Pass One preserves this exact separation rather than re-deriving it fuzzily.

In-Person Mic & Imported Files (Acoustic Diarisation):

For mono audio sources, Pass One uses an energy and zero-crossing rate voice activity detector (`detectSpeechSegments`) combined with acoustic spectral clustering (`inferSegmentSpeaker`) to segment speech turns.

4. Low-Confidence Escalation Policy

A mumbled name, indistinct postcode, or quiet telephone audio is the primary failure mode of automated acoustic redaction.

4.1 Threshold & Escalation Rule

Threshold: `CONFIDENCE_THRESHOLD = 0.70`.

Handling Policy: Words with confidence score ≤ 0.70 are marked with `isLowConfidence: true` and `escalateToAdviserReview: true`.

Zero-Drop Rule: Low-confidence regions are never silently dropped. They are catalogued in `LocalAsrResult.lowConfidenceWords` and escalated directly to the Phase 9 Adviser Redaction Verification Gate for mandatory manual sign-off.

5. Memory Management & Lifecycle Cleanup

All Pass One data structures are held within `VolatileSessionStore`:

`VolatileSessionStore.localAsrResult`: Holds structured segments, word timestamps, and confidence scores.

`VolatileSessionStore.localDraftTranscript`: String representation for internal pipeline review.

On session destruction (`destroySession()`), session reset, or consent withdrawal, all references are cleared and typed arrays zero-filled.